

```

        printf("Sent %s\n" to client\n", buffer);
    }

    /* Initiate SHUTDOWN sequence. Use SO_LINGER
    * socket option (via setsockopt) to send ABORTs
    */
    close(new_fd);
}
}

```

And here's the code for *client.c*:

```

#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>

#include <sys/types.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <arpa/inet.h>
#include <netinet.h>

int main()
{
    int sock_fd, ret, len, msg_flags;

    struct sockaddr_in serv_addr;
    struct sctp_event_subscribe event;
    struct sctp_sndrcvinfo sri;

    char buf[256];

    serv_addr.sin_family = AF_INET;
    serv_addr.sin_port = htons (2000);
    serv_addr.sin_addr.s_addr = inet_addr("127.0.0.1");

    sock_fd = socket (AF_INET, SOCK_STREAM, IPPROTO_SCTP);
    if (-1 == sock_fd)
    {
        perror("socket");
        exit(1);
    }

    if (connect (sock_fd, (struct sockaddr *) &serv_addr,
        sizeof(serv_addr)) < 0)
    {
        perror("connect");
        exit(1);
    }
    else
    {
        printf ("Unconnected to [%s]\n", inet_ntoa(serv_addr.sin_addr));
    }

    /* Not subscribing for any events. Not interested. */

```

```

memset (&event, 0, sizeof(event));
if (setsockopt(sock_fd, IPPROTO_SCTP, SCTP_EVENTS, &event, sizeof(event)) != 0)
{
    perror("setsockopt");
    return (-1);
}

sri.sinfo_stream = 1; /* Send data on stream no 1 */

/* Prepare data to be sent to server */
memset (buf, 0, sizeof(buf));
strcpy(buf, "Hello, SCTP world!");
ret = strlen(buf);

/* Send data to server */
len = sizeof(serv_addr);
ret = sctp_sendmsg (sock_fd, buf, ret, (sockaddr*)&serv_addr, len, 0, 0,
sri.sinfo_stream, 0, 0);
if (ret < 0)
{
    perror("sctp_sendmsg");
    return (-1);
}
else
{
    printf("Sent %s\n" to server\n", buf);
}

/* Receive what server sends back */
ret = sctp_rcvmsg (sock_fd, buf, sizeof(buf), 0, 0, &sri, &msg_flags);
if (ret < 0)
{
    perror("sctp_rcvmsg");
}

printf ("Received message from server on stream %d: %s\n", sri.sinfo_stream, buf);

close (sock_fd);

return 0;
}

```

I hope walking you through this simple client-server program gives you an idea about the virtues of SCTP.

Further reading

- Man pages: <http://linux.die.net/man/7/sctp>
- SCTP specific structure details: <http://www.kernel-api.org> 

By: Srikanta Prasanna.

The author likes his beard and handmade khadi clothes. In his leisure time, he works for Nokia Siemens Networks, Bangalore. You can reach him via his Gmail ID srikantap.